Antibiotics

1. Azithromycin
2. Ciprofloxacin
3. Levofloxacin
4. Cefixime
5. Ceftriaxone
6. Cefotaxime
7. Ceftazidime
8. Cefuroxime
9. Cefepime
10. Meropenem
11. Imipenem
12. Piperacillin tazobactam
13. Vancomycin
14. Clindamycin
15. clarithromycin
16. Erythromycin
17. Moxifloxacin
18. Ofloxacin
19. Norfloxacin
20. Cefpodoxime-proxetil
21. Cefdinir
22. Cefaclor
23. Cefoperazone
24. Ceftibuten
25. Cefoxitin
26. Cefotetan
27. Cefpirome
28. Doripenem

29. Ertapenem
30. Teicoplanin
31. Tobramycin
32. Netilmicin
33. Rifampicin
34. Rifabutin
35. Rifaximin
36. Fosfomycin
37. Fusidic acid
38. Gatifloxacin
39. Gemifloxacin
40. Minocycline
41. Oxytetracycline
42. Ticarcillin
43. Ticarcillin + Clavulanic acid
44. Piperacillin
45. Roxithromycin
46. Spiramycin
47. Josamycin
48. Kanamycin
49. Streptomycin
50. Pristinamycin

Micro-organism name	Resistance cases
1. Streptococcus pneumoniae	
2. Escherichia coli	26402
3. Streptococcus pneumoniae	
4. Escherichia coli	
5. Escherichia coli	16360
6. Escherichia coli	26212
7. Pseudomonas aeruginosa	
8. Haemophilus influenzae	
9. Pseudomonas aeruginosa	
10. Klebsiella pneumoniae	5327
11. Pseudomonas aeruginosa	7369
12. Pseudomonas aeruginosa	13600
13. Staphylococcus aureus	
14. Staphylococcus aureus	2547
15. Streptococcus pneumoniae	
16. Staphylococcus aureus	3999
17. Streptococcus pneumoniae	
18. Escherichia coli	
19. Escherichia coli	
20. Escherichia coli	
21. Streptococcus pneumoniae	
22. Haemophilus influenzae	
23. Pseudomonas aeruginosa	
24. Escherichia coli	
25. Bacteroides fragilis	
26. Bacteroides fragilis	
27. Staphylococcus aureus	
28. Pseudomonas aeruginosa	

29. Escherichia coli

30. Staphylococcus aureus

31. Pseudomonas aeruginosa
32. Escherichia coli
33. Mycobacterium tuberculosis

34. Mycobacterium avium

35. Escherichia coli

36. Escherichia coli
37. Staphylococcus aureus
38. Streptococcus pneumoniae
39. Streptococcus pneumoniae
40. Staphylococcus aureus
41. Escherichia coli
42. Pseudomonas aeruginosa

43. Pseudomonas aeruginosa

44. Pseudomonas aeruginosa
45. Streptococcus pneumoniae
46. Streptococcus pyogenes
47. Staphylococcus aureus
48. Escherichia coli
49. Mycobacterium tuberculosis

50. Staphylococcus aureus

Resistance gene	diagnostic test
erm(B), erm(A), mef(A/E), mph(A), msr(D)	Kirby-Bauer Disc Dffusion
gyrA, gyrB, parC, parE mutations; qnrA/B/S, aac(6')-Ib-cr, qepA	Kirby-Bauer Disc Dffusion
gyrA, gyrB, parC, parE; qnr genes	Kirby-Bauer Disc Dffusion
blaCTX-M, blaTEM, blaSHV, blaCMY	Disc Dffusion
blaCTX-M, blaSHV, blaTEM, blaCMY, blaDHA	Disc Dffusion
blaCTX-M, blaTEM, blaSHV, blaPER	Disc Dffusion
blaCTX-M, blaVEB, blaPER, blaGES, blaCMY	Disc Dffusion
blaTEM, blaSHV, blaCTX-M, blaOXA	Disc Dffusion
blaCTX-M, blaCMY, blaDHA, blaKPC, blaNDM	Disc Dffusion
blaKPC, blaNDM, blaVIM, blaIMP, blaOXA-48-like, blaOXA-23	Disc Dffusion
blaKPC, blaNDM, blaVIM, blaIMP, blaOXA-48-like, blaOXA-23, OprD loss	Disc Dffusion
blaCTX-M, blaTEM, blaSHV, blaOXA-1, blaCMY, blaDHA	Disc Dffusion
vanA, vanB, vanC, vanD, vanM	Broth microdiution
erm(A), erm(B), erm(C), lnu(A), lnu(B), vga genes	Disc Dffusion
erm(B), erm(A), mef(A/E), mph(A)	Disc Dffusion
erm(B), erm(A), mef(A/E), msr(D)	Disc Dffusion
gyrA, parC, gyrB, parE	Disc Dffusion
gyrA, parC, qnrA/B/S, aac(6')-Ib-cr	Disc Dffusion
gyrA, parC, qnrA/B/S, qepA	Disc Dffusion
blaCTX-M, blaTEM, blaSHV	Disc Dffusion
blaCTX-M, blaTEM, blaSHV	Disc Dffusion
blaTEM, blaSHV, blaCTX-M	Disc Dffusion
blaCTX-M, blaSHV, blaCMY, blaNDM	Disc Dffusion
blaCTX-M, blaTEM, blaSHV	Disc Dffusion
blaCMY, blaDHA, blaACT, blaFOX; mecA/mecC in S. aureus	Disc Dffusion
blaCMY, blaDHA, blaACT, blaFOX	Disc Dffusion
blaCTX-M, blaSHV, blaCMY, blaKPC	Disc Dffusion
blaKPC, blaNDM, blaVIM, blaIMP, blaOXA-48-like	valideted disc

blaKPC, blaNDM, blaVIM, blaIMP, blaOXA-48-like; porin loss with ESBL/AmpC	Disc Dffusion
vanA, vanB, vanC, vanD, vanM	Broth microdiution
aac(6')-Ib, ant(2'')-Ia, aph(3')-VI, rmt, armA	Disc Dffusion
aac(6')-Ib, ant(2'')-Ia, aph(3')	Disc Dffusion
rpoB mutations	MIC / Disc Dffusion
rpoB mutations; altered drug susceptibility associated with specific mutations	MIC / Disc Dffusion
rpoB mutations	MIC
fosA, fosA3, fosB, fosC2, glpT, uhpT, murA mutations	Agar diution
fusB, fusC, fusD, fusA mutations	Disc Dffusion
gyrA, parC, gyrB, parE	Disc Dffusion
gyrA, parC, gyrB, parE	Disc Dffusion
tet(K), tet(L), tet(M), tet(O), tet(S)	Disc Dffusion
tet(A), tet(B), tet(K), tet(L), tet(M), tet(O)	Disc Dffusion
blaTEM, blaSHV, blaOXA, blaPSE, blaCARB	Disc Dffusion
blaTEM, blaSHV, blaOXA, blaCTX-M, blaCMY	Disc Dffusion
blaTEM, blaSHV, blaCTX-M, blaOXA, blaCMY	Disc Dffusion
erm(B), erm(A), mef(A/E)	MIC / Vaideted disc diffusion
erm(B), erm(A), mef(A/E)	MIC / Vaideted disc diffusion
erm(B), erm(A), mef(A/E)	MIC / Vaideted disc diffusion
aph(3')-IIIa, aph(3')-VI, ant(4')-Ia, rmt	Disc Dffusion
strA, strB, aadA, rpsL, rrs mutations	Disc Dffusion
erm(A), erm(B), erm(C), vat, vgb, vga, lsa, msr(A)	MIC / Vaideted disc diffusion

Aternative antibiotic

Doxycycline, minocycline, susceptible beta-lactams

TMP-SMX, nitrofurantoin or fosfomycin
beta-lactams, doxycycline or TMP-SMX
cefepime, carbapenems
Carbapenems for serious ESBL
Carbapenems or susceptible alternative beta-lactams
Ceftazidime-avibactam, cefiderocol
Susceptible beta-lactams
Carbapenems; newer beta-lactam combinations

Ceftazidime-avibactam, meropenem-vaborbactam, aztreonam-avibactam, cefiderocol, depending on organism and enzyme

Ceftazidime-avibactam, imipenem-relebactam, cefiderocol or other active agents

Carbapenems for serious ESBL

Linezolid, daptomycin, tigecycline, depending on organism and infection
Doxycycline, minocycline, linezolid

Doxycycline, minocycline or other organism-specific agents

Doxycycline, minocycline, susceptible beta-lactams
Doxycycline or other organism-specific alternatives
beta-lactams, TMP-SMX
Nitrofurantoin, fosfomycin or TMP-SMX
oral beta-lactams or other suitable oral agents
Susceptible oral alternatives based on infection site
Susceptible beta-lactams
beta-lactams or carbapenems if appropriate
Susceptible oral beta-lactams
Carbapenems or other susceptible agents for Gram-negative organisms; MRSA-active agents for MRSA
Other susceptible beta-lactams

Susceptible carbapenems or newer beta-lactam combinations

Ceftazidime-avibactam, cefiderocol or other susceptibility-confirmed agents

Ceftazidime-avibactam, meropenem-vaborbactam or other mechanism-appropriate agents
Linezolid, daptomycin
Amikacin, gentamicin or non-aminoglycoside agents
Amikacin or other active agents
Depends on organism and infection

Depends on organism and infection

Alternative treatment depends on the indication and organism
Nitrofurantoin, TMP-SMX or other susceptible urinary agents
beta-lactams for MSSA, doxycycline, TMP-SMX
Susceptible organism-specific alternatives
Doxycycline or susceptible beta-lactams
Doxycycline, susceptible beta-lactams or linezolid
Doxycycline, minocycline
Piperacillin-tazobactam or other susceptible beta-lactams
Other beta-lactam/beta-lactamase inhibitor combinations, guided by AST
Piperacillin-tazobactam or other active beta-lactams
Doxycycline, minocycline
Organism-specific susceptible alternatives
Doxycycline, minocycline or susceptible alternatives
Amikacin or other susceptible agents
Susceptibility-guided non-aminoglycoside agents

Linezolid, daptomycin, or other susceptible Gram-positive agents